

Question:
A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

- Answer Choices:
- A. Epstein-Barr virus (EBV)
 - B. Measles virus
 - C. Human immunodeficiency virus (HIV)
 - D. Human papillomavirus

Reasoning + Answer:

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CD21 is a receptor on B cells that functions as a co-receptor in the recognition of antigens, particularly in the context of B cell receptor signaling. It is part of the B cell receptor complex and plays a critical role in the enhancement of B cell activation, especially in response to antigens presented by complement proteins. CD21 binds to the complement protein C3d, which is found on certain pathogens. This interaction helps in the efficient recognition and activation of B cells in response to antigens on pathogens.

Among the listed pathogens:

- Epstein-Barr virus (EBV): EBV infects B cells and interacts with B cell receptors. However, the role of CD21 is not central to the EBV pathogenesis; instead, EBV relies on B cell proliferation and growth signaling.

- Measles virus: The measles virus utilizes the complement system and is known to interact with complement components. The detection of measles virus antigens by B cells through the C3d/CD21 pathway is crucial for the formation of neutralizing antibodies. Loss of CD21 would impair the B cell response to measles virus, thereby affecting humoral immunity against measles.

- HIV: HIV infects CD4+ T cells and macrophages and does not directly interact with CD21 or B cell receptors.

- Human papillomavirus (HPV): HPV primarily infects epithelial cells and does not significantly depend on B cell signaling or CD21.

Thus, the pathogenesis of measles virus is most likely affected by the genetic modification that eliminates CD21 expression, because CD21-mediated recognition of complement-coated measles antigens is vital for effective B cell activation and anti-measles antibody production.

< answer > Measles virus </answer >

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